For an Autonomous Car Braking System, the safety stakes are among the highest in engineering. Unlike a human driver who might "feel" a brake fade, an AI depends entirely on sensor data and electronic actuators.

1. Identify the Hazard

Hazard: Unintended Acceleration or Failure to Decelerate.

Potential for Harm: High-speed collision with obstacles, other vehicles, or pedestrians, resulting in fatalities and significant property damage.

2. Risk Assessment Matrix

In the automotive industry, we use ISO 26262 standards to determine the ASIL (Automotive Safety Integrity Level). Braking is almost always ASIL D—the highest level of rigor.

Probability	Negligible	Marginal	Critical
Frequent (City Driving)	Medium	High	Extreme
Probable	Low	Medium	High
Occasional	Low	Medium	High
Remote	Low	Low	Medium

Assessment: Because a vehicle spends thousands of hours on the road (High Exposure) and the driver often cannot intervene in time (Low Controllability), any failure in the braking system is classified as Extreme risk.

3. Safety Requirement

This requirement focuses on "fail-operational" or "fail-safe" behavior.Requirement: "The system shall not inhibit braking commands from the Perception Layer when an object is detected within the Critical Safety Envelope (d < d_safe), regardless of software state."